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Generation of Non Invasive Measurement Signals for Digital Twin Based Knee Joint Deterioration Simulating of Drilling Employers

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Abstract

Drilling employers' knee joints deteriorate while influencing and developing by high pressure underground works, with undemand health issues influence productivity and downtime. Intermediate signal for simulating deterioration processes is necessary for helping health monitoring and work planning. Using acoustic emission(AE) as testing material, this study leverages generative AI agents and innovative prompting to generate simulation data, to explore signal generation and pattern classification of healthy and osteoarthritis(OA) knee signals among various age levels. The study is divided into two parts: AI driven knee AE generation(Ex-1) and pattern classification of (Ex-2). Ex-1 verifies the generation of healthy and OA simulation signals. A benchmark dataset consists of 53 knee joint AE signals, categorised by age and health status into 5 groups, namely, healthy-young(H1), healthy-middle-aged(H2), healthy-elderly(H3), OA-middle-aged(OA1), OA-elderly(OA2), are used for simulation. Through prompting, the GENAI-agent produces 20 signal sets per group, as well as corresponding novel designed visual based profiles(HD), and the autonomous pattern classification workflow could be done by using AIGC in order to verify the capability.

The proposed approach for AE detection in knee joint inspection shows that AIGC generated AE hits closely match actual data, except H3, differing by ~100 hits. Ratios for others range 0.8–1.2. Generated AE waveforms highly resemble measured ones, with correlation coefficients 0.76–0.99 and standard deviations <0.1. Euclidean distance based classification achieves ~97% accuracy, indicating potential for digital twin reconstruction of knee biomechanical deterioration via pre-trained models and prompt engineering. For pattern classification, GENAI-agent outperforms GENAI, exceeding 90% accuracy in OA1/OA2 diagnosis versus GENAI's 80%, demonstrating feasibility in occupational health management with minimal model modification. These findings enabling missing data compensation and high-precision, full-life cycle digital twin generation for managing occupational health risks for O&G employers. With integration of proposed approach into existed digital twin system, it is possible to help decision makers to make even comprehensive decisions.

Keywords: Generative AI, Digital twin, Signal generation, Pattern classification, Employers' health risks management

Introduction

Complex mechanism based equipment or bio-structures, such as rotating, reciprocating structures or human knees, exist in various industrial scenarios and effectively support the safe and orderly operation of complex industrial systems involving both humans and devices (Hao et al., 2024). Non-destructive testing (NDT) (Kong et al., 2020; Uldry et al., 2024), as a crucial means for the life cycle health management of complex structures, has been applied across multiple fields, including complex equipment (Amirhossein et al., 2024; Chen et al., 2013) and human bodies (Mascaro et al., 2009; Shark et al., 2011; Shark et al., 2016), with all typical life cycle monitoring process remains similar, covering the entire life cycle including data acquisition, data representation, feature engineering, pattern recognition, pattern classification, as well as data intelligent driven decision supporting (Ryder et al., 2023; Beaudoin et al., 2024).

With the continuous improvement in terms of life quality among complex biomechanical structures and full process monitoring modules, although the data volume and evaluation dimensions of monitored objects have significantly increased, the amount of data available for continuous learning and fault simulation to achieve abnormal evolution is decreasing due to the scenarios of early, especially lack of abnormal as well as intermediate data that presents progressing between healthy and deterioration, which might influence the simulation of current digital twin system and planning information systems such as productivity and downtime, with un-demand healthy issues were not included. Additionally, traditional methods for achieving high-precision virtual-real reconstruction of complex biomechanical structural state evolution face issues such as high computational costs, complex modelling processes and human computer interactions difficulties, and insufficient methodologies in terms of simulation data generation (Xu et al., 2023), forming new bottlenecks for the monitoring, diagnosis, classification and life cycle management of complex occupational health issues such as O&G employer knee joints.

The era of generative AI simplifies existing human-AI interaction modes, by using content generation to instead justification based methodologies, while also providing more derivative data services based on imagination and reasoning, leveraging contextual understanding and event association capabilities (Xia et al., 2024; Ghafarollahi et al., 2024). Through prompt engineering constructed via thought chains, combined with the generative AI based agents abilities to think, plan, decompose complex tasks, execute, and refine to form the close loop, it can further unleash the potential of generative AI, possessing the capacity to handle multiple scenario tasks to a certain extent (Durante et al., 2024).

However, the quality and efficiency enhancements aided by AIGC are primarily limited to simple and daily used application scenarios such as question answering and querying, formal report writing etc. For higher value tasks with complex workflow generation, latent space data generation and intermediate data simulation, the following major bottlenecks remain: (1) The modelling of complex decisions and the corresponding problem solving processes are lack of effective and continuously learnable datasets for understanding and pre-training of foundation AIGC models. (2) Efficient and low-cost cold boosting methods are lacking for complex decision solving and deduction processes. (3) The programs for generating simulated data targeting latent and hypothesis spaces are complex and inefficient. (4) Unable to generate valued data describing the latent space by purely rely on the combination of foundation models and generative AI agents.

To address these bottlenecks, this manuscript explores a simple and economical method for high precision simulation data generation through innovative prompt engineering combined with innovative data iteration methods. Using acoustic emission (AE) signals as the detection medium, complex structural objects- knee joints from O&G drilling field employers are studied. With a small amount of historical data as the stimulating dataset, this research generates datasets describing the hypothesis space for knee joints based

on innovative prompt engineering combined with iterative data processing method, and performs similarity assessments and validations of multi-level pattern classification tasks, thereby verifying the feasibility of approach proposed. The manuscript is organised as follows: Section 2 presents the experimental objectives and methods, including the acquisition, generation, and evaluation protocols of AE signals for O&G drilling employers' knee joints; the generation of pattern recognition test signals and task validation methods; and the design of prompt templates. Section 3 analyses the accuracy of AE signal generation and pattern classification for the studying objects. Conclusion and discussion is given in section 4.

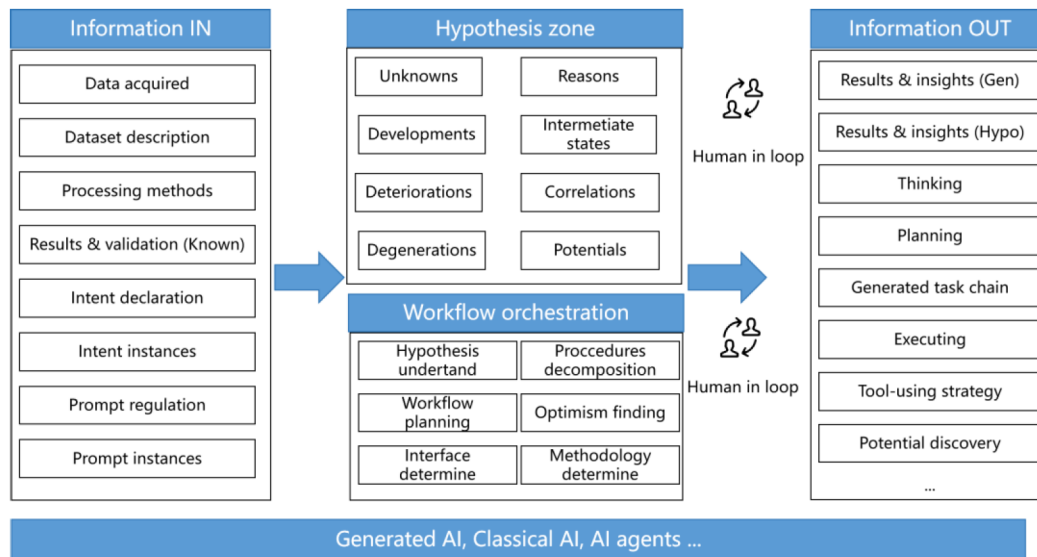


Figure 1—Procedures regarding the purpose of study

Methodology

The manuscript proposed a novel signal generation and simulation design based on the combination of prompt engineering and generative AI as tool-user to engage various data acquisition systems, as well as to integration multiple modality of data, thereby further reducing the entrance of non-invasive signal generation of O&G employers' knee conditions for intermediate progressing assessment. Utilising a small amount of physical acquired knee AE and joint angular signals, and guided by domain knowledge, the O&G employers' knee joint signals under different hypothesis and progress can be created by using the allied tool orchestration by Generative AI.

Generation of knee biomedical signals based on AIGC

The simulation of knee biomedical AE signals among O&G drilling employers (i.e. study objects) divided into two parts, namely, validation of the AE signal generation (Ex-1) and the ability to classify tool invocation patterns through generative AI (Ex-2), respectively. Ex-1 initially verifies the capability to generate healthy and OA simulation test signals, followed by the generation of signals representing intermediate states of degradation, with five groups of joint deterioration isolated by ages and symptoms. Model performance is determined through similarity comparisons with reference benchmark. The reference benchmark described in (Shark et al., 2011) is used as a reference, comprising AE signals collected from 53 knee joints based on 10-15 iterative sit-to-stand (STS) protocol (Mascaro et al., 2009), with instance illustrated in fig. 2 along with corresponding clinical diagnoses. This includes questionnaires and pathology to obtain subjects' lifestyles, past medical histories (ligament/meniscus injuries, knee injuries, surgical histories, chronic conditions), and OA status (age, gender, BMI, quantitative estimates of quality of life, occupation, exercise habits, medication habits). These data modules were taken into account, in order to

stimulate genAI for simulation data creation. To lower down the computing consumption, the foundation model use the document as the attach file without SFT employed to the foundation model.

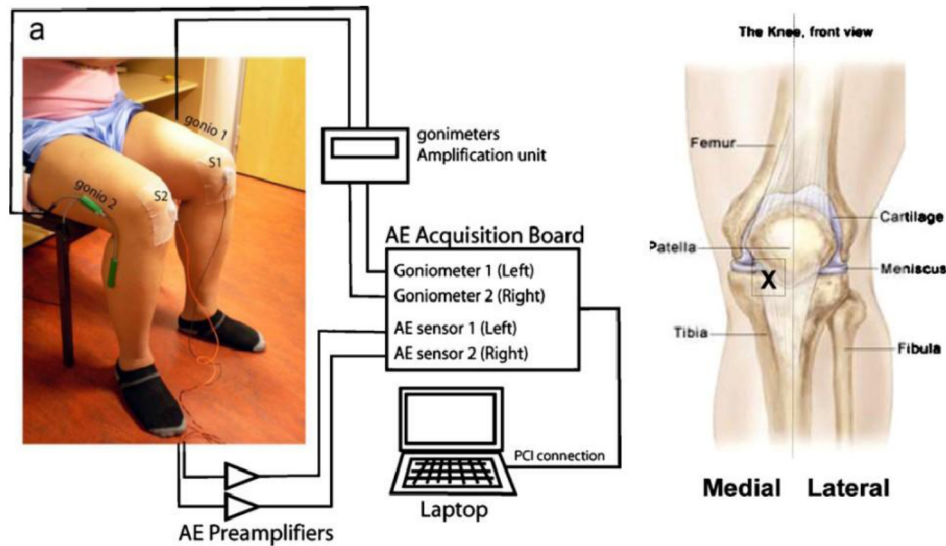


Figure 2—Illustration of STS protocol, with physical AE signal generated during testing subjects provide STS from a seat without armrest, cross arms in front of chest, and with piezo-electric sensors as well as goniometers attached on front and lateral of knee.

Based on this comprehensive information, a comprehensive knee joint assessment knowledge base is constructed. Subjects within the benchmark database are categorised by age and knee health status into: healthy-young group (H1), healthy-middle-aged group (H2), healthy-elderly group (H3), osteoarthritis-middle-aged and younger elderly group (OA1), and osteoarthritis-elderly group (OA2). The feasibility of these groupings has been verified in (Shark et al., 2011). All calibration data have been approved for use as experimental references. In terms of signal generation strategies, the aforementioned calibrated dataset and knowledge base serve as external resources. Through prompt engineering, the large model generates 20 signal sets for each of the five subject groups, with each AE signal sampled at a rate of 1 MHz, producing one AE hit every 15 ms and approximately 40 μV (32 dB). These 20 AE signal sets representing 20 digital twin based virtual drilling workers as the subjects being used for validation. Simultaneously, angular signals during the STS process are collected at an angular velocity of approximately 0.1°/s. As instance signals demonstrate Fig. 3, with AE signal acquired from OA knee have significant higher magnitude and average signal energy than healthy knee. During the experiment, comparisons include original AE signals and multimodal feature extraction methods, namely hits density per movement (G-HD), as first introduced in (Mascaro et al., 2009; Shark et al., 2011) to establish multimodal profiles, and to reduce the complexity of signal sets among five groups. The generated G-HDs are compared with the average profiles of the five reference groups (R-HDs) in terms of similarity and accuracy. A portion instances of G-HD profile generated by GenAI technology that representing five age groups of drilling employers' knee AE signals is illustrated in fig. 4, with pixel intensity and sub-fields represents significant differences over these five group divisions.

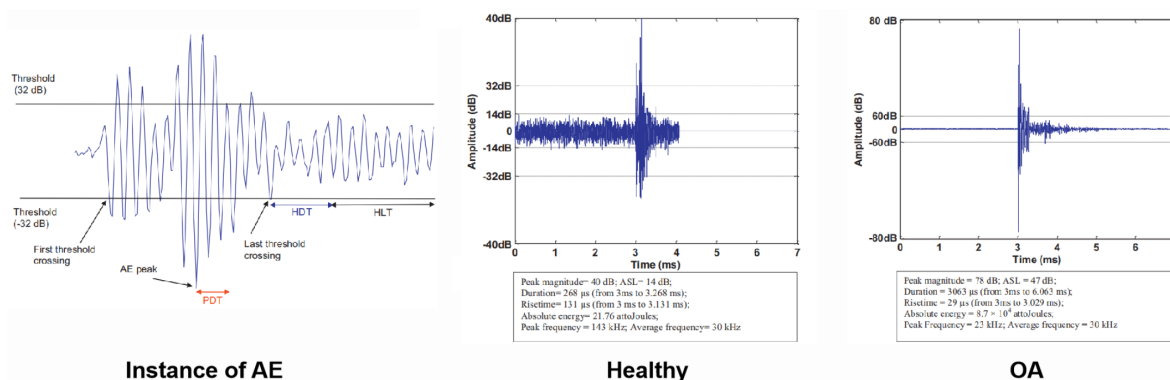


Figure 3—Instance of healthy and OA knee AE signal

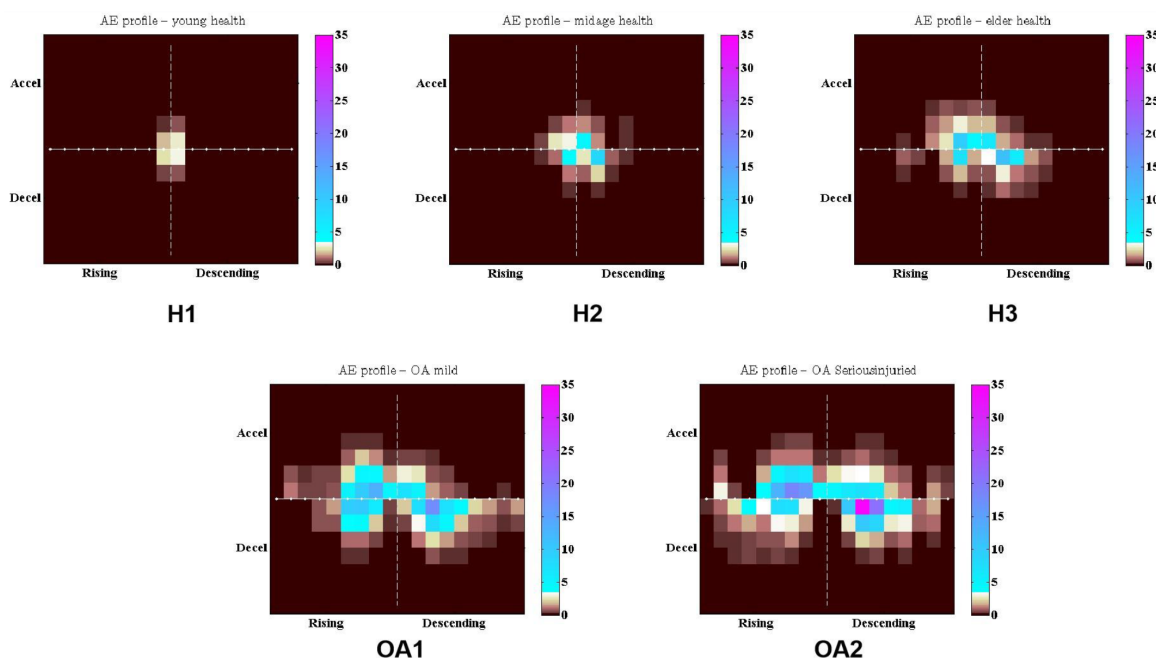


Figure 4—Instance of HD among drilling employers' knee among five age divisions

Ex-2 validates the method's capability for state signal generation as well as pattern classification ability under multiple constraint conditions. Eleven O&G industry frontline workers are selected as study subjects. Their actual knee joint conditions are diagnosed by professional medical students through questionnaires and physical/ultrasonography examinations, with clinical status and actual knee health profiles described in natural language. These diagnoses serve as the reference set for validation (i.e. to calibrate the reference centre, and to be used as the tag for vector distance calculation). Using the aforementioned prompt engineering and calibrated dataset as external resources, knee joint signals are generated based on natural language descriptions of health profiles (omitting the diagnosis results), and their health status and stages are determined based on these signals. Evaluations are conducted by using only generative AI model as well as reflexion agent framework (Shinn et al., 2023), respectively, by using the dimension of overall accuracy.

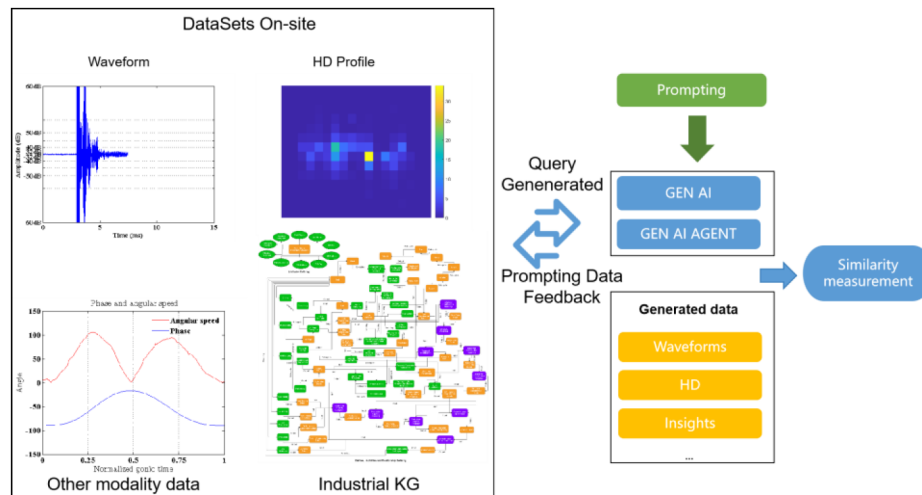


Figure 5—Experimental set-up

Two medium size LMs were selected as the foundation model for the above two experiments, serving as the generation and verification role, respectively, as well as to validate the accuracy of the generated signal dataset over various evaluating dimensions. During the pattern recognition phase acted by reflexion agent, a cumulative iteration strategy was employed for signal generation, where the generating role produced a dataset, which was then validated by the verification role based on predefined criteria. Reasonable results were included in the dataset, while unreasonable signals were discarded. Additionally, the verification role generated judgment criteria for discarded signals, which were incorporated into the subsequent generation strategy as a reference for the next iteration and new dataset generation. With all above mentioned categories applied, the continuous refining loop is formed with less human interference.

Prompting template

Based on experience and multiple experiments, the prompt template matching the aforementioned tasks should include:

- # *{General purpose}*: Confirm what LM or agent need to do.
- ## *{Role Definition}*: Clarify the role played by GENAI or GENAI-AGENT in the current scenario.
- ## *{Subject Definition}*: Describe the attributes and functional characteristics of the test subject.
- ## *{Task Definition}*: Clearly define the task for generating simulated signals.
- ## *{Generation Process Requirements}*: Declare the target data type and specify the requirements for the data to be generated, considering the application scenario.
- ## *{Reference Data Prompt}*: Declare the referenced dataset and the characteristics of the benchmark dataset.
- ## *{Reference Data Pattern Declaration}*: Provide natural language-based data feature descriptions for the target referenced data of GENAI, along with exemplary data samples and characteristic descriptions for specific states.
- ## *{Working Instructions}*: Prompt the LM on how to generate the dataset based on the chain-of-thought model, described in natural language. Optionally, provide individual examples through natural language descriptions.
- ## *{Correctness Judgment}*: Define correctness judgment rules in natural language, combining signal characteristics.

An instance of prompting for AE signals generation is demonstrated in Fig. 6:

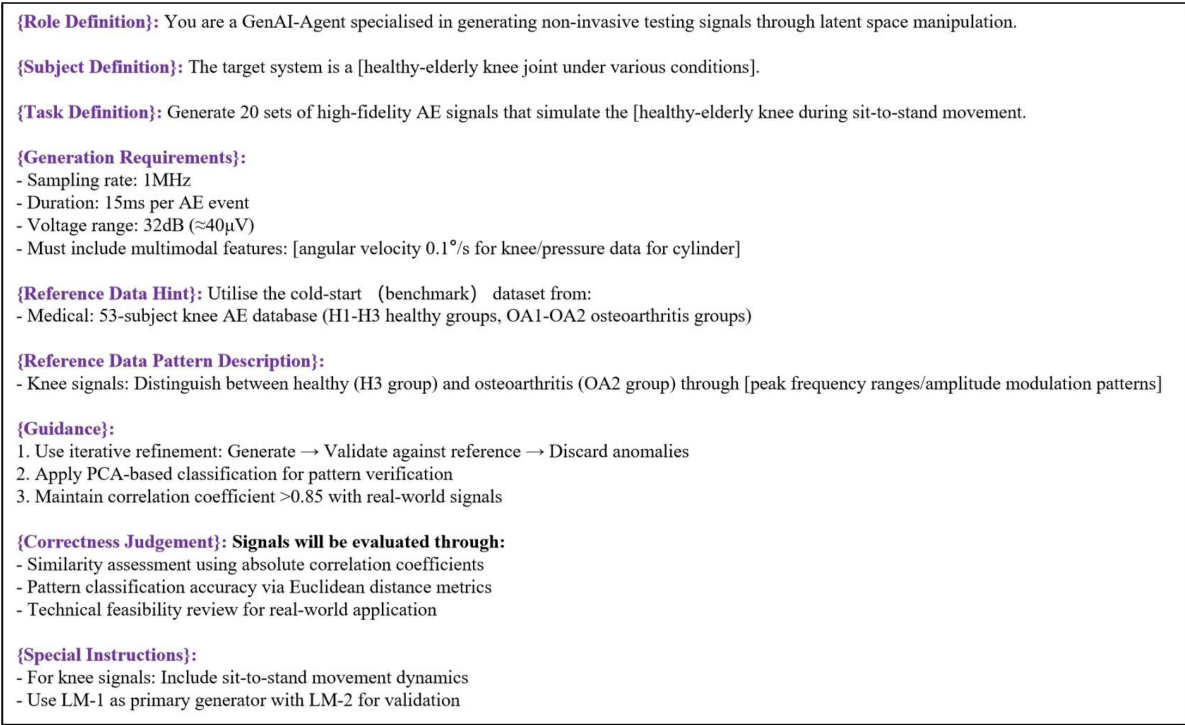


Figure 6—Instance of prompting created based on the prompting template

Results and discussion

The experimental results are presented in Table 1. Where, the reasonable AE features to describe knee conditions are:

1. Number of AE events: determine the number of AE events (AE hits) cross the threshold and be noted as a hit, with OA knees create significantly more AE hits than healthy (i.e. more OA more noisy than healthy). With amount of AE hits are considered as one of key feature to classify normal and abnormal, ratio of AE hits generated versus actual level that represent similarity between generated signals’ and physical acquired is worth to take into account to assess the performance of proposed approach;
2. Waveform similarity: to measure the similarity of AE signals created by AI with actual acquired by using correlation coefficient (CC) and standard deviation (SD). With differentiation described in previous section, some distinctive AE events only generated during deterioration process of knees.
3. Similarity of HD: to compare HD created by GENAI, with original HD of five groups, which provide apparent visual contrast among various deterioration levels or age groups.
4. Vector space distance: computation of the Euclidean distance through PCA projection, between the centre of original cluster and the projection of generated HD in the space formed by the first three principal eigenvectors;
5. Pattern classification using AIGC/agents: classification of generated signals, represented by HD, by using the optimal classification methodology autonomous selecting by GENAI or agents, with the accuracy determined by the consistency between their pre-defined knee class and the group classified.

Table 1—Evaluation the performance of AE signals created by GENAI and GENAI agents

Category	Healthy knees (H1/H2/H3) [average 5-10 mins for STS]	OA knees (OA1/OA2) [10-15 mins for STS OA]
Ratio for number of AE hits generated/ Actual	H1:77/83; H2:108/120 H3:256/356	OA1:352/311 OA2:560/483
Waveform similarity (Overall correlation in average absolute values)	(CC : 0.845, 0.772, 0.896) (SD : 0.032, 0.016, 0.005)	(CC : 0.863, 0.927) (SD : 0.069 0.047)
Hits density similarity	(CC:0.847,0.773,0.895) (SD : 0.027, 0.063, 0.012)	(CC:0.872, 0.941) (SD : 0.048, 0.009)
Similarity measurement through PCA projection (Classified by Euclean distance overall)	86.67%	87.50%
Pattern classification by using genAI (Overall accuracy)	77.27%	81.82%
Pattern classification by using genAI agents (Overall accuracy)	86.36%	90.91%

Based on the experimental results demonstrated in [Table 1](#), it is seen that as the primary indicator for AE detection, the AE hits generated by AIGC in study objects' knee joint inspection was generally close to those actually collected, with the exception of H3, which differs by approximately 100 hits. The ratios for the others are distributed between 0.8 and 1.2, which can be considered as "within acceptable level", as the level of AE hits among various groups are distinctive. In terms of the generated AE waveforms, by comparing the mean absolute values of the CC and the corresponding SD values, it is evident that the waveform generated by AIGC are highly similar to the actual waveforms acquired from the real knees. With the CC values range from 0.77 to 0.93, as well as SD values all below 0.1. Similarly, in terms of the performance of HD and its corresponding principal component analysis (PCA) projection, the mean values of CC are also distributed between 0.7 and 0.94, and with SD values all below 0.1. Based on classification results using Euclidean distance, the highest overall accuracy reaches 90.91%. A low percentage of error occurred during pattern classification, which is considered as normal issue, with the deterioration processes are progressing from H1 to OA2 as the nature.

These results suggest that through pre-training large models and simple prompt engineering, plus attached expert documentations, it is possible to address, to some extent, the issue of digital twin reconstruction and simulation for complex bio-mechanical targets in non-invasive testing under data-constrained conditions. Among the generated results, Group H3 exhibits relatively lower correlation and PCA projection classification accuracy. This phenomenon is strongly correlated with the intermediate state displayed by Group H3 between OA and healthy characteristics, leading to a relatively blurred similarity judgment based on data space compared to other results, albeit at a high level of similarity ([Mascaro et al., 2009](#); [Shark et al., 2011](#)), however, this is also reasonable that these intermediate states could provided early warning to the anomaly or may describe in detail the real situations regarding to heavy duty drilling employers.

Regarding the application of GENAI and GENAI-AGENT in pattern classification tasks, by projecting the HDs of the generated signals using PCA and then classifying them based on autonomous selected pattern classifier picked up by GENAI or GENAI agent, it is evident that using prompt templates as shown in this paper to generate prompt statements and guiding GENAI and GENAI-AGENT for pattern recognition and classification through these prompt statements demonstrates technical feasibility in workers' health diagnosis and management. Due to its abilities to think, plan, decompose tasks, execute tasks, and partially self-improve, GENAI-AGENT performs significantly better than GENAI in terms of pattern classification accuracy. In OA1/OA2 diagnosis state classification, its classification accuracy exceeds 90%, compared to an overall classification accuracy of 80% for GENAI, however all mis-classified patterns were incorrect

assigned between OA1 and OA2, but not mis-classified between Healthy and OA. This result indicates that the combination of precise prompt engineering and agent can greatly enhance the pattern classification capabilities of GENAI when facing complex structures and objects. Furthermore, in terms of model secondary development, the workload required to modify the model itself is minimal, providing a new method from the perspective of application economy, and the results also indicate that pre-warning could be made to emphasise the occupational health issues by reporting as slightly early state identification by following easy activities.

Conclusion

The manuscript presented a method for hidden layer space signal generation and pattern classification based on the integration of GENAI and prompt engineering innovations. Using 53 interpretable reference database and corresponding knowledge as external datasets, combined with novel designed thought chains constructed through prompt engineering, GENAI and generative AI agent demonstrate robust capabilities in simulated signal generation. Through experiments on the generation and pattern classification of AE signals from O&G drilling workers' knee AE signals, the feasibility of signal generated for description of structure evolution and pattern classifications are confirmed. This technique provides a feasible path for data generation in describing and evaluating complex structures in confined spaces and data-missing environments, as well as for pattern recognition and classification across various states of complex bio-mechanical structures. By using portion of data acquired from real knees, as guiding datasets and leveraging innovative prompt engineering techniques, AI generated data continuously enriches the dataset, especially enabling missing data compensation and high precision among latent space, which enables the full life cycle digital twin generation for non-invasive testing signals of complex bio-mechanical structures with minimal model fine-tuning and training costs. This study is current known first system for generative AI based digital twin signal generation for occupational health monitoring among O&G workers' knee, by whatever using this individually, or integrating it into the currently existed digital systems, it can be served as useful and novel tool for O&G employers' health risk prediction, to help reduce NPT, and improve quality of life among certain occupation.

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